Toward Zero-Forgetting: A Training-Free Multimodal Framework for Remote Sensing Class-Incremental Learning

Wen-Liang Du, *Member, IEEE*, Jicun He, Jiaqi Zhao, *Member, IEEE*, Rui Yao, *Member, IEEE*, Xiaoyi Li, and Yong Zhou

***Abstract*—Existing class-incremental learning (CIL) methods for remote sensing (RS) scene classification often tend to be training-intensive or rely on static visual features that may inadequately capture the complex interclass similarity and intraclass diversity inherent in RS imagery. Moreover, directly reusing features from models pretrained on natural images (e.g., CLIP) often suffers from a domain gap and limited semantic richness when transferred to the RS domain. In this article, we propose a simple yet effective training-free CIL framework for RS scene classification that leverages multimodal semantic information to build more discriminative category representations. Our framework treats pretrained models as frozen feature extractors to guarantee zero forgetting of the representation space. To enhance semantic discriminability, we employ a large-language model (LLM) to generate rich candidate textual descriptions for each class and introduce an image-guided description selection (IGDS) strategy to align semantic information with visual characteristics. Classification is performed using a distance-based metric without any additional training. Extensive experimental results demonstrate that our framework achieves leading performance and superior stability across different session sequences, surpassing both training-based and training-free baseline methods. Our code is available at https://github.com/WenliangDu/ZFCIL-RS**

***Index Terms*—Class-incremental learning (CIL), remote sensing (RS) scene classification, training-free.**

I. Introduction

REMOTE sensing (RS) scene classification serves as an essential foundation for numerous downstream applications [1], [2]. With the advancement of deep learning, impressive progress has been achieved on closed-set, static datasets. Nevertheless, real-world RS environments frequently introduce new categories dynamically through a stream format [3], [4], posing significant challenges for existing static models [5], [6], [7].

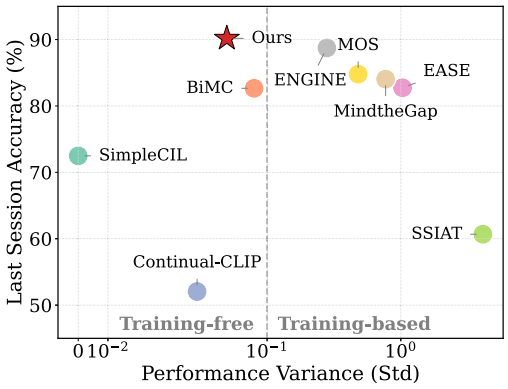


Fig. 1. Comparison of average accuracy against performance variance (standard deviation) for various incremental learning methods. Training-free methods generally exhibit lower variance, whereas training-based methods tend to achieve higher accuracy. Our framework achieves superior accuracy with minimal variance, outperforming both training-free and training-based baselines.

Deep learning-based class-incremental learning (CIL) is a technique designed to learn from new data continuously and build a unified classification model. Current CIL methods for RS scene classification can be roughly categorized into two types: the traditional training-from-scratch model and parameter-efficient tuning based on pretrained models. The former usually attempts to mitigate forgetting by constraining the weight updates or rehearsing old data through regularization and replay. The latter typically fixes the pretrained model to preserve its generalizability and introduces a limited number of trainable parameters. Despite these differences, both methodologies still rely on model training and therefore fail to overcome *catastrophic forgetting* and *overfitting* problems, resulting in significant prediction variances caused by different sequences in incremental learning (see training-based results in Fig. 1). This challenge prompts a critical question: *How can a training-free paradigm be developed that only treats the pretrained models as feature extractors and gives equal importance to historical and incremental data, thereby fundamentally addressing the issues of catastrophic forgetting and overfitting?*

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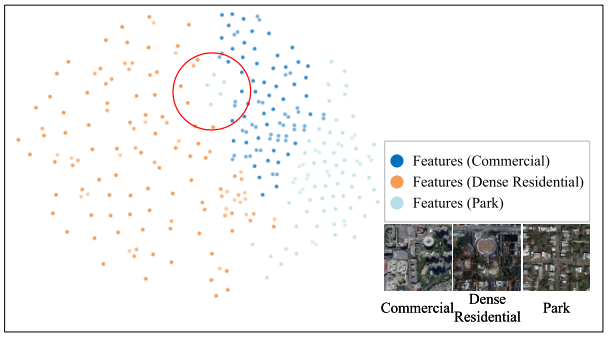


Fig. 2. RS scene classification often exhibits significant intraclass variance and interclass similarity, as illustrated by the red circles highlighting overlapping image features from different classes.

Training-free CIL emerges as a novel paradigm where the feature extractor, that is, pretrained models, remains frozen to guarantee zero forgetting of the representation space [8]. Pretrained models such as CLIP [9] can provide multimodal representations for both images and texts, enabling the alignment of image features with their corresponding class label descriptions. In this framework, the "learning" of a new class is reduced to calculating a prototype vector and registering it in the classifier. Classification is then performed by computing the similarity between the test image and all registered prototypes. However, RS scene classification is inherently challenging because scene categories are often characterized by global spatial layouts rather than salient local objects [6], [10]. In practice, RS images usually exhibit large intraclass variation due to differences in scale, viewpoint, illumination, seasonal conditions, and imaging sensors. Meanwhile, many categories share highly similar spectral and structural patterns, leading to strong interclass ambiguity. For example, visually similar scenes such as commercial areas, dense residential regions, and park areas (as shown in Fig. 2) may differ only in subtle contextual cues. These properties make RS CIL particularly vulnerable to prototype confusion and classification instability.

In this article, we propose a simple yet effective training-free CIL framework for RS scene classification. Our framework leverages pretrained models as the feature extraction backbone, fusing features extracted from images of the same category with their corresponding category information to form a multimodal prototype. To ensure that each prototype can comprehensively represent category features, thereby effectively distinguishing interclass similarity and intraclass diversity inherent in RS images, our approach not only utilizes CLIP to extract features from images and their corresponding category texts but also employs a large language model (LLM) to generate richer and more discriminative semantic descriptions [11]. Specifically, we propose an image-guided description selection (IGDS) module that filters descriptions generated by the LLM based on their correlation with the image representation, effectively selecting the most relevant descriptions guided by the image content. This process effectively reduces the impact of LLM hallucinations. Furthermore, we design an incremental multimodal prototype construction (IMPC) module that integrates image, text, and selected description features into a unified multimodal prototype. Finally, a prototype pool is established to serve as the reference set, enabling conventional machine learning classifiers to classify test images based on these representative prototypes without requiring any gradient-based learning.

Our main contributions are summarized as follows.

1) To the best of our knowledge, this is the first training-free CIL framework proposed for RS scene classification that ensures zero forgetting of the representation space and treats historical and incremental RS categories with equal importance.
2) Our framework integrates features extracted from images, text, and filtered descriptions to create a comprehensive multimodal prototype, addressing the high intraclass variance and interclass similarity commonly found in RS imagery.
3) Extensive experimental results on four RS CIL scene classification benchmarks demonstrate the superiority of the proposed simple but effective framework, which achieves state-of-the-art (SOTA) performance.

II. Related Work

A. Training-Based CIL

CIL requires a model to adapt to new categories while retaining knowledge of previously learned ones [12]. Most mainstream CIL methods achieve this by continuously modifying and updating model parameters when learning new classes. Training-based methods can be further divided into the traditional training-from-scratch models and parameter-efficient tuning based on pretrained models. Traditional training-from-scratch methods can be broadly categorized into four types: replay-based methods [13], [14], [15], [16], [17], [18] that store exemplars from old classes to preserve knowledge, regularization-based methods [19], [20], [21], [22] that impose constraints on parameter updates to mitigate forgetting, dynamic-network-based methods [23], [24], [25] that adapt network architecture for new classes, and knowledge-distillation-based methods [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36] that transfer dark knowledge from previous models to preserve representations. However, because training-from-scratch methods often involve substantial model updates whenever new knowledge is learned, they struggle to maintain stable classification performance. Pretrained-model-based methods offer a new solution by leveraging the strong representation capability learned from large-scale datasets [9]. The dominant pretrained-model-based approaches include prompt-based methods [37], [38], [39] and adapter-based methods [40], [41], [42]. These methods preserve the model's general representation ability while fine-tuning only a small number of parameters to accommodate new classes. In addition, with the rapid progress of vision–language models [9], approaches that fine-tune parameters and leverage textual guidance for visual classification have also achieved strong performance [43], [44], [45]. Recent parameter-efficient methods such as FlyPrompt [39] and BOFA [46] improve the adaptation of pretrained models in continual learning while mitigating forgetting. FlyPrompt uses

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a brain-inspired router and temporal ensemble for training-based adaptation, whereas BOFA restricts updates to CLIP's bridge layer with orthogonal low-rank fusion and hybrid prototypes, also remaining training-based. However, due to the significant domain gap between natural images and RS imagery [10], directly applying VLMs pretrained on natural image–text pairs often leads to limited semantic richness and suboptimal representation for RS scene classification.

B. Training-Free CIL

Parameter updates inevitably overwrite old knowledge. Therefore, training-free methods are of significant research value. At present, research on training-free CIL remains limited. Motivated by CLIP's strong zero-shot classification capability, continual-CLIP [47] extends CLIP into a continual learning framework and serves as a foundational model for training-free research. SimpleCIL [8] adopts a simple training-free paradigm by storing the mean image embeddings extracted from a fully frozen pretrained model in a classifier. In the realm of few-shot CIL, there has been a significant proliferation of training-free approaches. The work in [5], [48], [49], and [50] introduced training-free prototype calibration methods that enhance the discriminability of novel classes by calibrating prototype representations, while others [51], [52] specifically addressed the few-shot scenario in SAR target recognition through scattering-aware feature augmentation. Meanwhile, BiMC [11] implemented a bilevel modal calibration strategy within a training-free framework, demonstrating substantial efficacy in both few-shot and standard CIL scenarios. HyCal [53] extends training-free prototype calibration for cross-discipline few-shot CIL by combining cosine similarity and Mahalanobis distance on frozen CLIP embeddings to create stable, drift-resistant prototypes without parameter updates. Furthermore, Kang et al. [54] achieved a training-free paradigm by substituting the gradient update process with a generative approach based on diffusion models trained on base-session images. Despite these advances, existing training-free methods still face challenges in RS scenarios due to the unique characteristics of aerial and satellite imagery, including complex texture patterns, varying imaging conditions, and subtle interclass differences. Building upon these methodologies, we propose a training-free framework tailored to RS scene classification. By leveraging LLM-generated semantic descriptions and image-guided selection, our method is designed to better handle the high interclass similarity and intraclass variability commonly observed in RS scenes. Compared with existing training-free methods, our framework introduces richer multimodal class representations while preserving the simplicity and stability of a training-free pipeline.

III. Methodology

A. Overview

CIL aims to enable a model to learn from a sequence of sessions arriving sequentially. Consider a sequence of T sessions with the overall dataset as $\mathcal{D} = \mathcal{D}^1, \mathcal{D}^2, \ldots, \mathcal{D}^T$, where each session-specific dataset is given by $\mathcal{D}^t = (x_i, y_i)_{i=1}^{n^t}$. Here, n^t represents the total number of instances in the tth session, $x_i \in \mathcal{X}^t$ is an instance from the input space of the tth session, and $y_i \in \mathcal{Y}^t$ is its corresponding label. The label spaces $\mathcal{Y}^t$ are mutually disjoint, that is, $\mathcal{Y}^t \cap \mathcal{Y}^{t'} = \emptyset$ for all $t \neq t'$. After completing the Tth session, the model is expected to exhibit robust predictive performance on a test set that includes all classes from the entire sequence of sessions, encompassing both previous and current classes.

In our training-free setting, learning a new class is formulated as constructing and registering a representative prototype rather than updating model parameters. For each class, the images provide an image prototype, while the class label provides two forms of semantic information: a class-name text feature and a set of LLM-generated descriptions. These image and semantic cues are progressively organized into a unified multimodal prototype, which serves as the class representation during incremental inference.

To achieve this, we conduct two principal modules: **IGDS** and **IMPC** (as shown in Fig. 3). The IGDS module uses the image prototype as visual guidance to select the most relevant LLM-generated description features and aggregates them into a description prototype. The resulting description prototype is then passed to the IMPC module, where it is fused with the image prototype and the class-name text feature to form a unified multimodal prototype. As new classes arrive, their multimodal prototypes are incrementally added to a prototype pool, which is used for training-free classification through a distance-based classifier.

B. Image-Guided Description Selection

CLIP relies on simple and static prompt templates that cannot effectively capture the diverse visual characteristics of RS images, which often exhibit high intraclass variance and interclass similarity issues. To address this, we use an LLM to generate dynamic, category-specific descriptions, enabling the model to acquire more discriminative feature representations across RS categories. Furthermore, to eliminate description-generation errors caused by LLM hallucinations, we use the image information extracted by the frozen CLIP text encoder to filter and validate LLM-generated descriptions, ensuring more accurate and relevant category representations.

Specifically, for each class l, all training images belonging to this class are first encoded by the frozen CLIP image encoder, and their mean feature is used as the image prototype $\mathbf{p}_l$. This image prototype provides a visual summary of class l and serves as the guidance signal for selecting semantically consistent descriptions.

To obtain description features, an LLM is used to generate m sentences by enriching the corresponding scene category name. The prompt we employ is as follows: *"You are an expert in RS scene classification. For the RS scene category "[CLASS_NAME]," please generate {m} diverse and detailed textual descriptions that capture the distinctive visual characteristics, texture patterns, color distributions, and spatial structures typically associated with this category in aerial or satellite imagery. Each description should highlight unique discriminative features that help distinguish [CLASS_NAME] from similar scene categories."*

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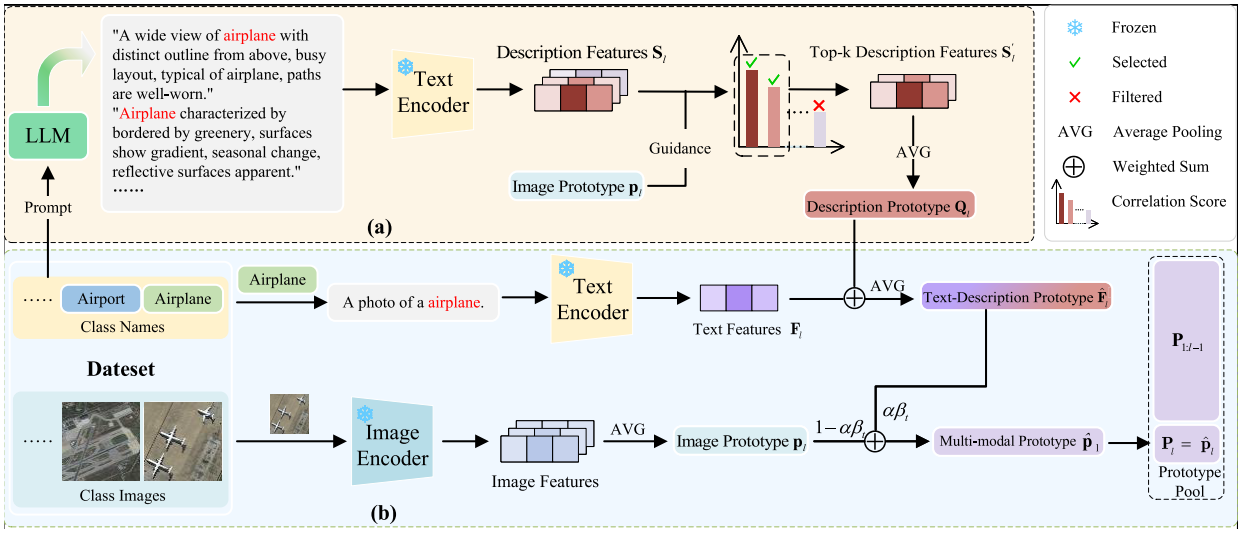


Fig. 3. Overall pipeline of the proposed multimodal prototype construction framework. (a) IGDS module generates remote-sensing-specific descriptions with an LLM and selects visually relevant descriptions according to their correlation with image prototypes. (b) IMPC module explicitly takes the image prototype, class-name text feature, and selected description features as inputs, and fuses them into a unified prototype stored in the prototype pool for training-free classification.

These sentences are encoded using the CLIP text encoder to produce m description features $\mathbf{S}_l = \{\mathbf{S}_{l,j}\}_{j=1}^{m}$, where $\mathbf{s}_{l,j}$ denote the jth description feature in $\mathbf{S}_l$.

Next, we select the top-k description features that exhibit the highest correlation with the image prototype $\mathbf{p}_l$. The selection is based on the Pearson correlation coefficient between each description feature and the image prototype. First, both the description feature and the image prototype are zero-centered as

$$\tilde{\mathbf{s}}_{l,j} = \mathbf{s}_{l,j} - \bar{\mathbf{s}}_{l,j} \tag{1}$$

$$\tilde{\mathbf{p}}_l = \mathbf{p}_l - \bar{\mathbf{p}}_l \tag{2}$$

where $\tilde{\mathbf{s}}_{l,j}$ and $\tilde{\mathbf{p}}_l$ denote the zero-centered vectors of $\mathbf{s}_{l,j}$ and $\mathbf{p}_l$, obtained by subtracting their respective mean values $\bar{\mathbf{s}}_{l,j}$ and $\bar{\mathbf{p}}_l$.

Then, the Pearson correlation coefficient $\rho_{l,j}$ between $\tilde{\mathbf{s}}_{l,j}$ and $\tilde{\mathbf{p}}_l$ is computed as

$$\rho_{l,j} = \frac{\operatorname{cov}\left(\tilde{\mathbf{p}}_l, \tilde{\mathbf{s}}_{l,j}\right)}{\sigma\left(\tilde{\mathbf{p}}_l\right) \cdot \sigma\left(\tilde{\mathbf{s}}_{l,j}\right)} \tag{3}$$

where $\operatorname{cov}(\cdot)$ and $\sigma(\cdot)$ denote the covariance and standard deviation, respectively.

Finally, the top-k description features $\mathbf{S}'_l$ are selected based on the k highest $\rho_{l,j}$ values, and then the description prototype $\mathbf{Q}_l$ is computed by averaging $\mathbf{S}'_l$. The obtained $\mathbf{Q}_l$ summarizes the visually relevant semantic descriptions of class l and is subsequently fed into the IMPC module for multimodal prototype construction.

C. Incremental Multimodal Prototype Construction

In this module, the information from images, text, and descriptions is fused to construct a multimodal prototype. First, the text and description information are combined into a text-description prototype. The text feature $\mathbf{F}_l$ for class l is extracted by feeding the prompt "a photograph of a [CLS]." into the frozen CLIP text encoder, where [CLS] is replaced by the class name. Then, $\mathbf{F}_l$ is combined with the description prototype $\mathbf{Q}_l$ from IGDS to form a text-description prototype $\hat{\mathbf{F}}_l$

$$\hat{\mathbf{F}}_l = \frac{\mathbf{F}_l + \mathbf{Q}_l}{2}. \tag{4}$$

Next, this text-description prototype is fused with the corresponding image prototype $\mathbf{p}_l$ to form the final multimodal prototype $\hat{\mathbf{p}}_l$

$$\hat{\mathbf{p}}_l = \alpha\beta_l \hat{\mathbf{F}}_l + (1 - \alpha\beta_l)\, \mathbf{p}_l \tag{5}$$

where α is a manually set hyperparameter, and β_l is a weighting factor computed as the similarity probability between the image prototype $\mathbf{p}_l$ and the description feature $\mathbf{Q}_l$

$$\beta_l = \operatorname{softmax}\left(\mathbf{p}_l \times \mathbf{Q}_l^{\top}\right). \tag{6}$$

Here, β_l adaptively controls the contribution of the semantic prototype according to its consistency with the image prototype. A higher similarity between $\mathbf{p}_l$ and $\mathbf{Q}_l$ indicates that the selected descriptions are more visually aligned with the class images, and thus their semantic information can be assigned a larger weight.

For the training dataset, a pool of multimodal prototypes $\mathbf{P} = \{\hat{\mathbf{p}}\}_{\mathbf{l=1}}^{\mathbf{L}}$ is constructed for each class across all the sessions. When a new session arrives, only the prototypes of newly observed classes are constructed and appended to this pool, while the existing prototypes remain unchanged. For the inference phase, we only extract the image feature $\mathbf{p}'$ for each test image, and the classification is performed by computing the Mahalanobis distance between $\mathbf{p}'$ and each class's multimodal prototype

$$M_l = \sqrt{(\mathbf{p}' - \hat{\mathbf{p}}_l)^{\top} \Sigma^{-1} (\mathbf{p}' - \hat{\mathbf{p}}_l)} \tag{7}$$

where Σ is the covariance matrix of the feature space, and M_l is the Mahalanobis distance of $\mathbf{p}'$ to the lth class's multimodal prototype $\hat{\mathbf{p}}_l$. The predicted class label C for the test image is assigned as

$$C = \arg\min_{1 \le l \le L} M_l. \tag{8}$$

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IV. Experiment

A. Implementation Details

We compare the proposed method with three training-free and eight training-based SOTA CIL methods. The training-free methods include one single-modal method (SimpleCIL [8]) and two vision–language model-based methods (Continual-CLIP [47] and BiMC [11]). The training-based CIL methods consist of five single-modal methods (CODA-Prompt [37], APER + Adapter [8], EASE [40], SSIAT [41], and MOS [42]) and two vision-language model-based methods (MindTheGap [44] and ENGINE [45]). All pretrained model backbones use the ViT-B/16 architecture.

Four RS scene classification datasets are used to evaluate the above CIL methods: Northwestern Polytechnical University (NWPU-RESISC45) [55], the UC-Merced land-use dataset (UC-Merced) [56], the Aerial Image dataset (AID) [57], and the RS Image Classification Benchmark (RSI-CB256) [58]. Specifically, NWPU-RESISC45 contains 31 500 images with a resolution of 256×256 pixels, covering 45 scene classes with 700 images per class; UC-Merced comprises 21 classes, each with 100 images, most of which have a resolution of 256×256 pixels; AID includes 10 000 images at 600×600 pixels, spanning 30 different scene categories with several hundred images per class; and RSI-CB256 consists of more than 24 000 images from 35 categories, each with several hundred images per class and sized 256×256 pixels. Following the ECA-MSDWNet method [59], we randomly split NWPU-RESISC45, UC-Merced, AID, and RSI-CB256 into five, three, six, and nine sessions, respectively, to evaluate the methods' capabilities.

All experiments are implemented on a server equipped with an NVIDIA 3090 GPU. We configure the hyperparameter α based on the highest training accuracy on each dataset. Hence, we set α = 0.7, 0.6, 0.04, and 0.6 for RSI-CB256, NWPU-RESISC45, UC-Merced, and AID datasets, respectively. We use GPT-5.2 for generating the category descriptions. We denote A_{last} as the accuracy of all classes in the last session, and A_{avg} as the average accuracy across all incremental sessions. To evaluate the stability with respect to different learning orders of sessions, we assess all models using three distinct session sequences and compare their mean A_{avg} and A_{last}, as well as the standard deviation across these orders. All compared methods are reproduced following their official source code settings.

We also record the computational overhead of the main steps for reproducibility. Across all datasets and random seeds, the complete experimental pipeline can be completed within 30 min, with a peak GPU memory usage of 2361.31 MB. The LLM-generated descriptions are obtained once in an offline class-level preprocessing stage and cached for later use, so they do not introduce additional runtime during prototype construction or inference. For prototype construction, the average CLIP image-encoding time per class ranges from 354 ms (on UC-Merced) to 1247 ms (on RSI-CB256). The Top-k selection over m generated descriptions and the subsequent multimodal prototype construction require 3.1 ms (on NWPU-RESISC45) to 4.8 ms per class (on UC-Merced) on average. For inference, the average processing time per test image ranges from 5.8 ms (on NWPU-RESISC45) to 12.6 ms (on UC-Merced).

B. Comparison Results

Table I shows the results of the proposed method compared with both training-free and training-based CIL methods. Obviously, the performance of training-based methods is generally higher than that of training-free methods, except for our framework, due to the additional training involved. However, the standard deviations of A_{last} for most training-based methods are higher than those of training-free methods. This phenomenon can be attributed to the nature of RS scene categories, which may exhibit similarities in the feature space. If the training order causes semantically similar categories to be distributed across different learning stages, the model tends to confuse new classes with old ones, leading to catastrophic forgetting. Additionally, when a session batch contains highly similar classes, the model struggles to distinguish between them, resulting in intrasession class conflicts. However, among the training-based methods, APER + Adapter is an exception, as its standard deviations for both A_{avg} and A_{last} are very low. This is because it employs a stored image prototype strategy and only performs parameter training during the initial session, without requiring updates during subsequent incremental sessions. Hence, APER + Adapter can also be regarded as a semi-training-based method. Our approach demonstrates competitive performance across all four datasets and significantly outperforms training-based methods without requiring any training, while also showing relatively low standard deviations compared to other training-free methods. This indicates that the training-free paradigm can achieve satisfactory results in CIL sessions for RS scene classification.

Specifically, on the NWPU-RESISC45 dataset, which is characterized by its large scale and high category complexity, our training-free framework achieves an impressive A_{last} of 90.20% and an A_{avg} of 93.56%. Compared to the leading training-free baseline, BiMC, our framework provides a substantial performance leap of 7.54% in the last session accuracy. More notably, it even surpasses the most advanced training-based models such as ENGINE, which achieves an A_{last} of 88.74%. This performance gap highlights that by effectively integrating LLM-generated semantic descriptions with image prototypes, our model can navigate the complex feature space of increasing RS scene classes more effectively than models that rely on parameter updates.

In the evaluation of the UC-Merced dataset, our approach reaches a peak A_{last} of 95.71% and an A_{avg} of 96.75%. When measured against other training-free strategies, our framework significantly exceeds the performance of SimpleCIL (90.71% A_{last}) and BiMC (89.76% A_{last}). Furthermore, our training-free paradigm outperforms sophisticated training-based methods like SSIAT and MOS, which reach 93.89% and 94.20% in A_{last}, respectively. The results on this dataset are particularly revealing because, despite the relatively small size of UC-Merced, our framework maintains a remarkably low standard deviation, whereas most training-based methods suffer from increased variance as the learning order changes.

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TABLE I

AVERAGE AND LAST PERFORMANCE COMPARISON ON FOUR DATASETS. THE BEST OVERALL PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND-BEST PERFORMANCE WITHIN DIFFERENT PARADIGM SETTINGS IS SHOWN UNDERLINED. ± REPRESENTS THE STANDARD DEVIATION

Method	Paradigm	NWPU-RESISC45		UC-Merced		AID		RSI-CB256	
		A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑
Continual-CLIP [47]	Training-free	$52.05_{\pm 0.04}$	$61.93_{\pm 0.75}$	$63.33_{\pm 0.00}$	$63.13_{\pm 7.19}$	$55.65_{\pm 0.04}$	$67.26_{\pm 3.58}$	$34.89_{\pm 0.02}$	$47.91_{\pm 2.93}$
SimpleCIL [8]		$72.48_{\pm 0.00}$	$79.19_{\pm 0.19}$	$\underline{90.71}_{\pm 0.00}$	$\underline{92.74}_{\pm 0.93}$	$81.90_{\pm 0.00}$	$87.08_{\pm 0.91}$	$\underline{92.34}_{\pm 0.00}$	$\underline{94.79}_{\pm 0.31}$
BiMC [11]		$\underline{82.66}_{\pm 0.08}$	$\underline{88.51}_{\pm 0.19}$	$89.76_{\pm 0.34}$	$91.23_{\pm 2.67}$	$\underline{89.15}_{\pm 0.53}$	$\underline{93.31}_{\pm 0.29}$	$83.45_{\pm 0.13}$	$89.93_{\pm 1.11}$
CODA-prompt [37]	Training-based	$79.95_{\pm 1.07}$	$88.44_{\pm 0.82}$	$86.35_{\pm 0.82}$	$92.16_{\pm 2.02}$	$71.98_{\pm 2.39}$	$82.18_{\pm 1.74}$	$78.90_{\pm 1.97}$	$87.66_{\pm 0.53}$
APER+Adapter [8]		$78.18_{\pm 0.64}$	$84.36_{\pm 0.75}$	$90.71_{\pm 0.00}$	$92.78_{\pm 0.96}$	$82.23_{\pm 0.25}$	$87.36_{\pm 1.06}$	$92.51_{\pm 0.08}$	$94.98_{\pm 0.28}$
EASE [40]		$82.74_{\pm 1.03}$	$88.78_{\pm 1.61}$	$89.45_{\pm 0.22}$	$92.04_{\pm 1.23}$	$80.33_{\pm 0.92}$	$86.35_{\pm 1.09}$	$91.53_{\pm 1.04}$	$95.10_{\pm 0.78}$
SSIAT [41]		$60.68_{\pm 4.11}$	$77.11_{\pm 0.91}$	$93.89_{\pm 1.41}$	$95.70_{\pm 0.59}$	$76.90_{\pm 2.63}$	$86.43_{\pm 1.66}$	$81.94_{\pm 5.39}$	$92.12_{\pm 0.60}$
MOS [42]		$84.79_{\pm 0.48}$	$91.28_{\pm 0.74}$	$94.20_{\pm 1.51}$	$96.24_{\pm 0.71}$	$88.65_{\pm 0.35}$	$92.55_{\pm 0.29}$	$94.69_{\pm 0.64}$	$96.86_{\pm 0.44}$
MindtheGap [44]		$84.04_{\pm 0.77}$	$90.84_{\pm 1.09}$	$89.36_{\pm 0.79}$	$90.74_{\pm 3.79}$	$83.15_{\pm 0.82}$	$91.11_{\pm 0.99}$	$71.80_{\pm 8.08}$	$96.42_{\pm 2.72}$
ENGINE [45]		$\underline{88.74}_{\pm 0.28}$	$\underline{92.94}_{\pm 0.33}$	$\underline{94.76}_{\pm 0.39}$	$\underline{96.35}_{\pm 1.13}$	$\underline{95.63}_{\pm 0.16}$	$\underline{97.34}_{\pm 0.18}$	$\underline{95.59}_{\pm 0.04}$	$\underline{97.71}_{\pm 0.16}$
FlyPrompt [39]		$82.64_{\pm 0.23}$	$87.06_{\pm 0.81}$	$77.38_{\pm 0.89}$	$82.26_{\pm 2.20}$	$78.92_{\pm 2.08}$	$78.39_{\pm 1.03}$	$93.98_{\pm 1.66}$	$95.26_{\pm 0.85}$
Ours	Training-free	$\mathbf{90.20}_{\pm 0.02}$	$\mathbf{93.56}_{\pm 0.31}$	$\mathbf{95.71}_{\pm 0.19}$	$\mathbf{96.75}_{\pm 1.39}$	$\mathbf{96.25}_{\pm 0.08}$	$\mathbf{97.35}_{\pm 0.29}$	$\mathbf{97.00}_{\pm 0.15}$	$\mathbf{97.90}_{\pm 0.20}$

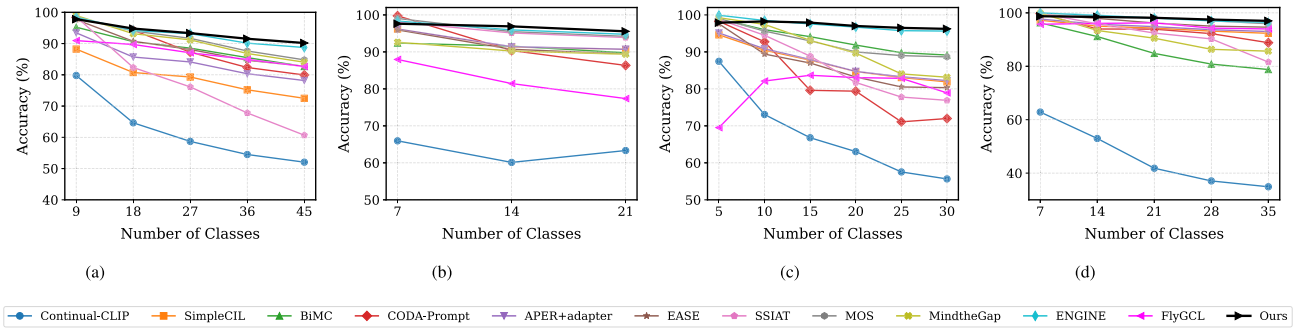


Fig. 4. Performance of each learning session on four datasets. (a) NWPU-RESISC45, (b) UC-Merced, (c) AID, and (d) RSI-CB256.

On the AID benchmark, the strengths of our IGDS module become even more apparent. Our framework achieves an A_{last} of 96.25% and an A_{avg} of 97.35%. This average accuracy slightly outperforms the SOTA training-based method, ENGINE, which records 97.34%. The performance margin over training-free baselines is even wider, with our A_{last} exceeding that of BiMC by 7.1%. This indicates that our framework's ability to filter LLM hallucinations and construct compact multimodal prototypes allows it to handle the high intraclass variance typical of aerial imagery without the need for computationally expensive gradient-based learning.

Finally, on the RSI-CB256 dataset, our framework delivers its most robust results, securing an A_{last} of 97.00% and an A_{avg} of 97.90%. This reflects an improvement of 4.66% in A_{last} over the best training-free baseline, SimpleCIL. In the competitive landscape of training-based methods, our approach still maintains a clear advantage over top-performing models like MOS (94.69% A_{last}) and ENGINE (95.59% A_{last}). A critical observation across all datasets, but especially on RSI-CB256, is that our framework exhibits the lowest standard deviation among all competitive paradigms. Unlike training-based methods, where standard deviations for A_{last} are high due to catastrophic forgetting and sensitivity to class order, our training-free strategy ensures that the representation space remains stable, thereby fundamentally addressing the stability-plasticity dilemma in RS incremental sessions.

We analyze the performance curves across all learning sessions as illustrated in Fig. 4, to further evaluate the stability and incremental learning capabilities of our framework. Across all four datasets, most methods experience a performance decline as the task sequence progresses; our proposed method maintains the highest and most stable accuracy throughout the entire incremental process. Most training-based methods (e.g., SSIAT and CODA-Prompt) often suffer from significant fluctuations and steep declines in accuracy during intermediate sessions. This is primarily due to catastrophic forgetting, where the optimization for new tasks inevitably disrupts the internal representations of previously learned categories. Some training-based methods (e.g., APER + Adapter and EASE) exhibit relatively stable performance, but they still fall short of our framework's accuracy, particularly in the later sessions. This is because these methods rely on parameter updates that can be sensitive to the order of class introduction and the inherent complexity of RS images. A notable observation is the performance of FlyPrompt on the AID dataset, which demonstrates relatively low accuracy in the first session, followed by an upward trend in accuracy across the subsequent two sessions. This unstable result suggests that FlyPrompt is not well-suited for high-resolution RS images, which exhibit greater intraclass variance and interclass similarity issues.

Training-free methods often lack the discriminative power necessary for complex RS imagery. For instance, continual-CLIP shows severe degradation and substantial fluctuations during incremental tasks because it relies on static prompts that fail to capture the high intraclass variance of RS scenes. Similarly, while SimpleCIL provides a stable baseline, its performance is significantly lower on datasets like AID and NWPU-RESISC45 compared to our multimodal approach.

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TABLE II

AVERAGE (A_{avg}) AND THE LAST (A_{last}) ACCURACY COMPARISON ON FOUR NATURAL IMAGE CLASSIFICATION DATASETS. THE BEST RESULTS ON EACH DATASET ARE INDICATED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. ± REPRESENTS THE STANDARD DEVIATION OF A_{last}. ↑ MEANS HIGHER IS BETTER

Method	CIFAR100		Cars		CUB		ObjectNet	
	A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑	A_{last} ↑	A_{avg} ↑
Continual-CLIP [47]	$66.66_{\pm 0.02}$	75.26	$58.08_{\pm 0.01}$	71.21	$54.00_{\pm 0.02}$	66.85	$44.07_{\pm 0.01}$	55.26
SimpleCIL [8]	$\underline{76.21}_{\pm 0.00}$	<u>81.66</u>	$41.69_{\pm 0.00}$	52.96	$\mathbf{85.24}_{\pm \mathbf{0.00}}$	**91.31**	$51.13_{\pm 0.00}$	61.25
BiMC [11]	$73.63_{\pm 0.06}$	80.75	$\underline{75.45}_{\pm 0.12}$	<u>84.18</u>	$76.79_{\pm 0.05}$	85.41	$\underline{54.51}_{\pm 0.10}$	<u>65.3</u>
Ours	$\mathbf{76.46}_{\pm 0.06}$	**83.19**	$\mathbf{79.38}_{\pm 0.02}$	**86.39**	$\underline{78.65}_{\pm 0.08}$	<u>86.37</u>	$\mathbf{58.46}_{\pm 0.14}$	**68.36**

TABLE III

BWT COMPARISON ON FOUR DATASETS. THE BEST RESULTS ON EACH DATASET ARE INDICATED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. BWT IS BETTER WHEN IT IS HIGHER

Method	Paradigm	NWPU-RESISC45	UC-Merced	AID	RSI-CB256
Continual-CLIP [47]	Training-free	−12.40	−3.01	−5.19	−9.90
SimpleCIL [8]		−7.67	−2.50	−6.85	−3.17
BiMC [11]		<u>−7.20</u>	−3.93	<u>−2.60</u>	−7.32
CODA-prompt [37]	Training-based	−9.97	−4.41	−13.85	−10.94
APER+Adapter [8]		−6.50	−2.62	−6.68	−2.88
EASE [40]		−7.40	−4.17	−9.98	<u>−2.62</u>
SSIAT [41]		−40.82	−3.30	−17.16	−16.06
MOS [42]		−8.92	−2.62	−5.44	−3.24
MindtheGap [44]		−7.45	<u>−2.06</u>	−6.91	−4.41
ENGINE [45]		−6.84	−3.21	−2.60	−3.33
FlyPrompt [39]		<u>−0.27</u>	**−0.16**	<u>−0.02</u>	**−0.01**
Ours	Training-free	**−0.21**	<u>−0.18</u>	**0.53**	<u>−0.03</u>

C. Evaluation on Natural Image Classification

To evaluate the generalizability of our framework beyond RS imagery, we conducted comparative experiments against the three training-free methods on four natural image classification benchmarks: CIFAR100 [60], the Stanford Cars dataset (Cars) [61], the Caltech-UCSD Birds-200–2011 dataset (CUB) [62], and ObjectNet [63]. Following established protocols, these datasets were split into 10, 10, 20, and 20 sessions, respectively.

As shown in Table II, our framework demonstrates competitive performance across all benchmarks. On the CIFAR-100 dataset, our framework attains the highest accuracy (A_{last} of 76.46% and A_{avg} of 83.19%). While SimpleCIL achieves the lowest standard deviation, our multimodal approach yields the best overall accuracy. For the Stanford Cars and ObjectNet datasets, our framework secures the highest accuracy and the lowest standard deviation on A_{avg}, reaching 79.38% A_{last} on Cars and 58.46% A_{last} on ObjectNet. Regarding the CUB-200 dataset, SimpleCIL delivers the top accuracy, with our framework achieving the second-best results (A_{last} of 78.65%).

In summary, while specifically optimized for the complex visual characteristics of RS, our framework maintains robust performance on general-purpose natural image tasks.

D. Evaluation on Forgetting

To further evaluate the ability of our framework to preserve historical knowledge while learning new categories, we conduct a detailed analysis of Backward Transfer (BWT) across all four RS benchmarks.

1) Definition and Formula: As defined in [64], BWT measures the influence that learning a new task has on the performance of previously learned tasks. A negative BWT value indicates catastrophic forgetting, where the acquisition of new information degrades the model's existing knowledge base. The mathematical formulation for BWT is given by

$$BWT = \frac{1}{T-1}\sum_{i=1}^{T-1} A_{T,i} - A_{i,i} \tag{9}$$

where T represents the total number of sessions, $A_{T,i}$ denotes the accuracy on the test set of the ith task after the model has completed learning up to the Tth task, and $A_{i,i}$ is the accuracy of the ith task immediately after it was learned.

2) Analysis of BWT Results: The comparative results for BWT are summarized in Table III. Our proposed framework demonstrates a competitive advantage over both training-based and existing training-free baselines.

Unlike most training-based traditional CIL methods that suffer from severe performance drops, such as SSIAT, which records a significant BWT of −40.82 on NWPU-RESISC45, FlyPrompt maintains BWT values exceptionally close to zero across all datasets. But the cost of this stability is a lower overall accuracy, particularly on the AID dataset, where FlyPrompt achieves only 78.39% average accuracy.

In contrast, our framework not only maintains a BWT close to zero but also achieves higher overall accuracy across all datasets. Specifically, on the NWPU-RESISC45 benchmark, our framework attains the best BWT of −0.21. On the AID dataset, we even observe a positive BWT of 0.53, suggesting that the accumulation of multimodal knowledge slightly enhances the representation of earlier classes. Although we only achieve the second-best results on the UC-Merced and RSI-CB256 benchmarks, the gap to FlyPrompt is merely 0.02, which still demonstrates the strong competitiveness of our framework.

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TABLE IV

ABLATION STUDY OF THE DIFFERENT MODULES IN OUR FRAMEWORK ON THE RSI-CB256 DATASET. WE REPORT THE OVERALL PERFORMANCE OF EACH SESSION AND THE AVERAGE PERFORMANCE. ALL EXPERIMENTS WERE CONDUCTED USING THREE RANDOM SEEDS, AND THE MEAN RESULTS ARE REPORTED

Ablation	Ses.1	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Avg↑
SimpleCIL	**99.37**	97.04	95.23	95.13	94.50	94.08	93.16	92.25	92.34	94.79
SimpleCIL+IMPC	93.70	98.70	98.35	98.23	**98.18**	**98.02**	97.34	96.84	94.39	97.08
SimpleCIL+IMPC+IGDS	97.85	**98.81**	**98.38**	**98.30**	98.16	97.97	**97.51**	**97.15**	**97.00**	**97.90**

TABLE V

ABLATION STUDY OF THE DIFFERENT DISTANCE METRICS IN OUR FRAMEWORK ON THE RSI-CB256 DATASET. WE REPORT THE OVERALL PERFORMANCE OF EACH SESSION AND THE AVERAGE PERFORMANCE. ALL EXPERIMENTS WERE CONDUCTED USING THREE RANDOM SEEDS, AND THE MEAN RESULTS ARE REPORTED

Distance Metric	Ses.1	Ses.2	Ses.3	Ses.4	Ses.5	Ses.6	Ses.7	Ses.8	Ses.9	Avg↑
Euclidean	69.02	59.12	36.49	56.16	32.13	17.91	18.66	65.31	49.18	44.89
Dot Product (SimpleCIL)	**99.83**	96.96	93.59	92.68	92.39	91.39	90.01	89.39	89.12	92.82
Manhattan	99.77	97.29	93.81	93.19	92.91	92.19	90.97	90.40	90.38	93.43
Mahalanobis (Ours)	97.85	**98.81**	**98.38**	**98.30**	**98.16**	**97.97**	**97.51**	**97.15**	**97.00**	**97.90**

These results confirm that by adopting a training-free paradigm with frozen backbones and multimodal prototype construction, our framework effectively eliminates catastrophic forgetting, ensuring that historical and incremental categories are preserved with equal integrity throughout the entire learning process.

E. Ablation Study

To evaluate the contribution of each component within our proposed framework, we conduct a series of ablation experiments on the RSI-CB256 dataset. All reported results are averaged over three random seeds to ensure statistical reliability. Notably, when our framework is configured to compute only the image prototypes and utilize these prototypes as the weights of a linear classifier, the architecture aligns with the SimpleCIL [8]. Consequently, we adopt SimpleCIL as the fundamental baseline for the ablation analyses.

Table IV presents the performance breakdown of our framework's core modules. While SimpleCIL achieves high accuracy in the initial task, it suffers from significant performance degradation as the number of incremental sessions increases, resulting in an average accuracy (A_{avg}) of 94.79%. Integrating the IMPC module to construct text–image prototypes and employing the Mahalanobis distance for classification significantly stabilizes incremental performance. Although it slightly underperforms the baseline in the initial task, it yields substantial gains in subsequent sessions, improving the final session accuracy to 94.39%. The integration of the IGDS module provides semantically rich category descriptions that are visually validated, achieving the best overall performance with an A_{avg} of 97.90%.

The choice of distance metric is critical for prototype-based classification in high-dimensional feature spaces. As shown in Table V, we compared the Mahalanobis distance used in our framework against three alternative metrics: Euclidean distance, Manhattan distance, and the dot product (the metric in SimpleCIL). The Euclidean distance yields poor performance in this context, with accuracy dropping below 20% in Sessions 6 and 7, highlighting its inability to handle the complex distribution of RS features. While the Manhattan distance performs competitively in the initial session, it fails to maintain stability as categories accumulate, reaching an average accuracy of 93.43%. The dot product achieves high initial session accuracy (99.83%), aligning with the high-performing characteristics of the SimpleCIL baseline. However, as the incremental sessions progress, its performance steadily declines to 89.12% in the final session, resulting in an average accuracy of 92.82%. Our framework, employing the Mahalanobis distance, achieves the best overall performance with an average accuracy of 97.90%. Although it begins slightly lower than the dot product in the first session, it demonstrates superior stability and accuracy in all subsequent incremental tasks, fundamentally mitigating the degradation seen in other metrics.

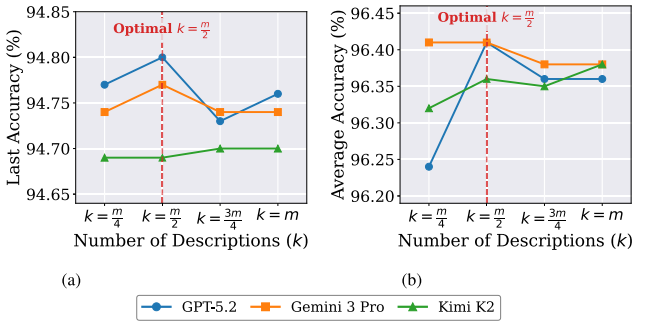


Fig. 5. Ablation study on the number of selected descriptions (K). (a) Shows the accuracy of the last incremental session, while (b) presents the average accuracy across all sessions. Results indicate that $K = m/2$ achieves the optimal tradeoff.

We further investigated the impact of the description selection parameter K and LLM choice. As illustrated in Fig. 5, we compared three different LLMs, GPT-5.2, Gemini 3 Pro, and Kimi K2, for generating scene descriptions. While all models exhibit high performance, they universally reach their optimal accuracy when the selection parameter is set to $K = m/2$. This trend underscores the effectiveness of our IGDS module in filtering redundant or "hallucinated" LLM content. In addition to classification accuracy, we also considered the practical accessibility and usage constraints of different LLMs under our experimental conditions. Since GPT-5.2 achieved performance comparable to Gemini 3 Pro and substantially better than Kimi

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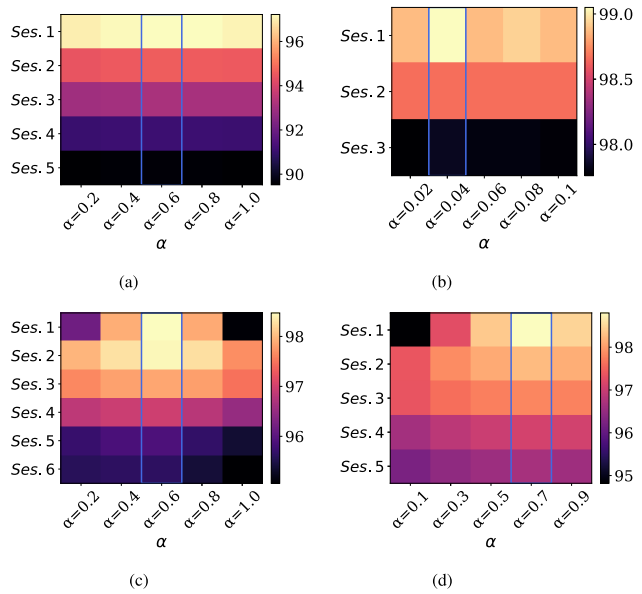


Fig. 6. Session-wise accuracy across different α values for the four RS datasets. (a) NWPU, (b) UCMerced, (c) AID, and (d) RSI-CB256.

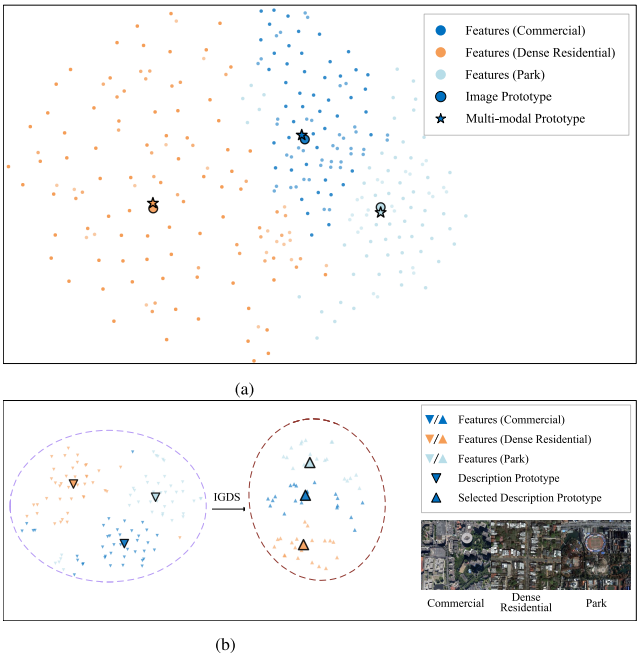


Fig. 7. T-SNE visualizations of image and description features on the AID dataset. (a) Image feature space of three semantically similar classes, together with the corresponding image-only and calibrated multimodal prototypes. The calibrated prototypes remain close to their associated image clusters, preserving visual consistency while enriching the class representation. (b) Description feature space before and after IGDS. The selected descriptions are more compact and separable, indicating that IGDS filters noisy semantic content and improves discriminability.

K2, while providing a favorable tradeoff in practical usability, we adopted GPT-5.2 as the default LLM for generating scene descriptions in our experiments.

The hyperparameter α regulates the fusion balance between visual image prototypes and LLM-generated semantic prototypes. Fig. 6 illustrates the session-wise accuracy across different α values for the four RS datasets. Obviously, the framework achieves peak performance on the RSI-CB256, NWPU, UCMerced, and AID datasets with α set to 0.7, 0.6, 0.04, and 0.6, respectively. Although the optimal α varies across datasets due to their different visual-semantic characteristics, the proposed framework is not overly sensitive to this hyperparameter. For instance, adopting a unified setting of $\alpha = 0.6$ across all datasets still leads to competitive results. Under this setting, UC-Merced achieves an A_{last} of 95.08 and an A_{avg} of 96.10, while RSI-CB256 obtains an A_{last} of 96.63 and an A_{avg} of 97.68. Therefore, $\alpha = 0.6$ can serve as a reasonable default setting in practice, while dataset-specific ablation is recommended when pursuing the best possible performance.

F. Visualization

To provide an intuitive view of the learned representation, we visualize three semantically similar categories from the AID dataset, namely commercial, dense residential, and park, using t-SNE [65]. As shown in Fig. 7(a), the image features of these categories exhibit a certain degree of overlap, which can be attributed to their shared spatial structures and spectral characteristics. The corresponding image-only prototypes are located near the feature regions of their respective categories, but the proximity between different class regions makes the classification task challenging. Following multimodal calibration, the calibrated prototypes (asterisk) remain close to their corresponding image-only prototypes (circle). This indicates that the proposed IMPC module performs a restrained prototype refinement rather than substantially changing the original visual representation. In this way, the introduced semantic information complements the image-based evidence while preserving the visual anchors used for classification. Such consistency is particularly important in the training-free CIL setting, where the frozen feature extractor is not updated, and the prediction is directly determined by the distances between image features and class prototypes.

Fig. 7(b) visualizes the textual feature distribution before and after applying the IGDS module. Without description selection, the raw semantic features generated by the LLM are scattered and partially overlapped across categories, indicating that some descriptions are redundant, noisy, or weakly aligned with the corresponding RS semantics. After IGDS, the class-wise textual features become more compact, showing that the image-guided selection strategy effectively filters out less informative content and retains the descriptions that are relevant to each class. This leads to more discriminative and precise semantic representations, which are beneficial for multimodal prototype construction in RS CIL.

Therefore, the visualization suggests that the proposed method enriches the prototype representation with multimodal information while preserving a stable visual correspondence, which helps reduce the ambiguity caused by the high interclass similarity and intraclass variability in RS scene classification.

V. Conclusion

In this article, we presented the first training-free CIL framework tailored for RS scene classification. Our research was motivated by the limitations of current training-based

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CIL methods, which suffer from overfitting and the loss of historical knowledge due to continuous parameter updates. By adopting a training-free paradigm, we ensure that historical and incremental categories are treated with equal importance, fundamentally addressing the issue of catastrophic forgetting.

By integrating visual features from a frozen vision–language model with LLM-generated semantic descriptions, our framework constructs a robust and expressive prototype representation for each class. The proposed IGDS mechanism further enhances semantic alignment, enabling the model to better capture interclass distinctions and intraclass diversity inherent in RS scenes. Importantly, the entire framework operates without parameter updates, rehearsal buffers, or session-specific retraining, making it both efficient and scalable.

Experimental results across multiple benchmarks and incremental settings demonstrate that our approach not only improves average accuracy but also significantly reduces performance variance compared to existing methods. This stability is particularly valuable in real-world RS systems, where reliability over long-term deployment is critical.

Although our framework achieves high classification performance with short runtime and low computational cost, its accuracy upper bound is still limited by the capability of the pretrained model, given its training-free nature. Therefore, in future work, we plan to build upon this foundation to explore how to obtain better image and text encoding models with minimal training cost, aiming to optimize the multimodal semantic fusion performance.

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